Phase 7: Project Documentation

1. System Overview

Rising Waters is an automated flood prediction platform. This documentation serves as a guide for both end-users and future developers.

2. End-User Manual

How to Use the Application

1. Navigate to the deployed web URL.
2. On the main dashboard, locate the input form.
3. Enter the required meteorological data:
   - **Annual Rainfall:** Total rainfall in mm.
   - **Cloud Visibility:** Percentage of cloud cover.
   - **Temperature:** Average temperature in Celsius.
   - **Humidity:** Relative humidity percentage.
   - **Seasonal Rainfall:** Monsoonal rainfall in mm.
4. Click **Predict**.
5. The system will output a "Flood" or "No Flood" alert, alongside a probability percentage.

3. Developer Guide

Local Environment Setup

To run this project locally, ensure you have Python 3 installed. ```bash

Install dependencies

pip install -r requirements.txt

(Optional) Re-train the model if data changes

python train_model.py

Start the Flask development server

python run.py `` `Access the application at`http://localhost:5000`.

Modifying the ML Model

To replace the XGBoost model with a different algorithm: 1. Modify `train_model.py`. 2. Ensure the new model is saved as a `.pkl` file in `app/models/`. 3. Update the loading logic in `app.py`.